Joonho Bae

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Academic Appointments

Kelley School of Business, Indiana University — Bloomington, IN
Assistant Professor of Operations and Decision Technologies — *Jan 2026 –*
Acting Assistant Professor of Operations and Decision Technologies — *July 2025 – Dec 2025*

Education

Stephen M. Ross School of Business, University of Michigan — Ann Arbor, MI
Ph.D. in Business Administration, Technology & Operations — *2026*
Committee: Roman Kapuscinski (Chair), John Silberholz, Eric Schwartz, and Xiuli Chao
Dissertation: Optimizing Battery Use for a More Sustainable Energy System

- Rackham Predoctoral Fellow (Highest research award at the University of Michigan)

Korea Advanced Institute of Science and Technology (KAIST) — South Korea
M.S. in Industrial & Systems Engineering — *2019*

Seoul National University (SNU) — South Korea
B.S. in Statistics, Minor in Financial Economics — *2016*

Research Interests

Topics: Sustainable operations; energy market; battery systems; risk management
Methodology: Optimization; dynamic programming; data-driven decision making

Research

Working Papers

1. Dynamic Operational versus Financial Hedging for a Risk-Averse Firm. [Link]
   - Co-authors: Roman Kapuscinski and Wanshan Zhu
   - Under review at **Management Science**

2. Dynamic Valuation of a Battery under Performance Degradation.
   - Co-authors: Roman Kapuscinski and John Silberholz
   - Preparing for submission

3. Joint Valuation of a Battery Portfolio: Benefits of Heterogeneity
   - Co-authors: Roman Kapuscinski and John Silberholz
   - Preparing for submission

4. Human-AI Cooperation or Competition? Optimize Data Sharing in Digital Markets. [Link]
   - Co-authors: Zijin Zhang
   - Finalist in INFORMS Service Science Cluster Best Paper 2026
   - In preparation for re-submission

5. No Label, No Problem? Retail Dynamics of Label-Free Bottles. [Link]
   - Co-authors: Yeolim Bella Yoon and Yeohong Yoon
   - In preparation for re-submission

6. Shorter Workweeks Shifting Wallets: How a Sustainable Labor Policy Reshapes Consumer Spending.
   - Co-authors: Yeolim Bella Yoon and Yeohong Yoon
   - Preparing for submission

7. Contract Design and Operations for Residential Battery Programs
   - Co-authors: Jerry Anunrojwong and Ali Fattahi
   - Preparing for submission

Published Papers

8. Cost-Saving Synergy: Energy Stacking in Battery Energy Storage Systems [Link]
   - Co-authors: Roman Kapuscinski and John Silberholz.
   - **Management Science**, 72(5), 3906–3924, 2026.

9. A Recommending System for Mobile Games using the Dynamic Nonparametric Model [Link]
   - Co-authors: Jeonghye Choi, Jinkyoo Park, and Seung-Bum Soh, 2023.
   - **Journal of Business Research**, 167, 114079, 2023.

10. Count-Based Change-Point Detection via Multi-Output Log-Gaussian Cox Processes [Link]
    - Co-author: Jinkyoo Park
    - **IISE Transactions** (formerly IIE Transactions), 52(9), 998–1013, 2020.
    - Preliminary version appeared at 2018 *Stochastic Processes and their Applications*, Gothenburg, Sweden

Work in Progress

11. Winds of Uncertainty: How Political Shocks Reshape Daily Life
    - Co-authors: Yeohong Yoon and Eunkyung An

Teaching Experience

Instructor – Indiana University Bloomington

Supply Chain Management with Digital Technologies (P 319, BBA; class size 30) *2026 – Present*
Business Analytics and Modeling (K 353, BBA; class size 38) *2026 – Present*

Instructor – University of Michigan

U-M Graduate Teacher Certificate [Link] *2022*
- Granted to individuals for their accomplished pedagogical training and experience as college-level instructors at the University of Michigan

Operations Management (TO 313, BBA Core; class size 79) *2021*
- Instructor evaluation: 4.9/5.0 [Link]
- Recipient, 2022 – 2023 Gerald and Lillian Dykstra Fellowship for Teaching Excellence (Top 1-2 graduate student instructors in Ross School of Business)

Teaching Assistant – University of Michigan (Graduate-Level Courses)

Applied Analytics for Decision Making (WMBA 623, avg. class size 45) *2022 – 2023*
Managerial Big Data Analytics (EMBA 610, class size 75) *2023, 2025*
Manufacturing and Supply Operations (TO 605, avg. class size 80) *2020 – 2023*
Analytics Primer (avg. class size 45) *2020 – 2023*
Applied Business Statistics (WMBA 503, class size 114) *2022*

Teaching Assistant – KAIST (Undergraduate-Level Courses)

Data-Driven Decision Making and Control (IE 481, class size 26) *2018*
Engineering Statistics I (IE 241, class size 50) *2018*

Professional Experience

Dean's Office, University of Michigan, ***Analytical Specialist*** *2023 – 2025*
- Assisted the Dean's Office at Ross School of Business with analytical projects

KAIST, ***Research Fellow*** *2019*
- System Intelligence Lab. (PI: Jinkyoo Park), Sponsors: **LG CNS & LG Chem**
- Developed real-time monitoring systems for battery energy storage systems (BESS)
- Implemented a data-driven approach to estimate state of health for BESS

SK Telecom, ***Marketing Manager*** *2015 – 2017*
- Managed more than 40 stores in Seoul, South Korea
- Developed optimizing tools for distribution of cell phones

Cham Son Gil Cooperative, ***Co-Founder & Project Leader*** *2013 – 2015*
- Established a healing center for blind masseurs and developed service/products targeting 20-30s, who were not the main customers of the original market
- Expanded branches nationwide (12 branches as of 2025), ensuring $3,000 monthly income for each masseur on average

Seoul National University, ***Research Assistant*** *2012 – 2013*
- Time Series Lab. (PI: Sinsub Cho)
- Revised the Korean Educational Statistics Software for statistical analysis in Excel

8th U.S. Army, ***Sergeant*** *2010 – 2012*
- Served as Korean Augmentation to the United States Army (KATUSA)
- Honorably discharged as sergeant (Army Commendation Medal)

Presentations (* denotes scheduled)

2026 POMS Annual Conference, INFORMS Annual Conference
2025 University of Toronto (Rotman Young Scholar Seminar), INFORMS Annual Conference

2024	POMS Annual Conference, KAIST, M&SOM Annual Conference, INFORMS Annual Conference, DSI Annual Conference
2023	Early-Career Sustainable Operations Workshop, POMS-HK Conference, POMS Annual Conference, M&SOM Annual Conference, INFORMS Annual Meeting
2022	M&SOM Annual Conference, INFORMS Annual Meeting
2021	INFORMS Annual Meeting
2018	Yonsei University Business School, Stochastic Processes and their Applications Conference, Global Marketing Conference, INFORMS Annual Meeting

Awards & Honors

Finalist in INFORMS Service Science Cluster Best Paper 2026 *2026*
University of Michigan, Rackham Graduate Student Research Grant *2025*
University of Michigan, Rackham Predoctoral Fellowship Award [Link] *2024 – 2025*
- Highest award for doctoral research at the University of Michigan

University of Michigan, Ross School of Business Research Merit Award *2024*
University of Michigan, Ross School of Business Fellowship *2019 – 2024*
Gerald and Lillian Dykstra Fellowship for Teaching Excellence *2022 – 2023*
University of Michigan, Ross School of Business Doctoral Research Grant *2023*
University of Michigan Graduate Teacher Certificate (GTC) *2022*
University of Michigan, Ross Early PhD Candidacy Award *2021*
KOSAF, National Scholarship (Science & Technology) *2009 – 2015*
Ministry of Employment & Labor of South Korea, Social Venture Idea *2014*
Enactus (Entrepreneurial Action Us), National Competition *2014*
Seoul Metropolitan Government, Hope Advertisement *2014*
Soonchunhyang University, SCH Social Venture Idea *2013*
SK Happiness Foundation, SK Social Enterprise *2013*

Service Activities

Ad hoc Reviewer, Management Science, Operations Research, Manufacturing & Service Operations Management, Production and Operations Management, Transportation Research Part E, Omega

Session Chair, POMS Annual Conference 2024 (Energy Markets and Storage), INFORMS Annual Meeting 2021 (Energy Sector)

University of Michigan Ross PhD Forum Research Committee Chair *2022 – 2023*
University of Michigan Ross PhD Forum Research Committee Member *2020 – 2023*
Cham Son Gil Cooperative Non-Profit Business Consulting *2015 – 2017*

Other Information

Optimization/data analysis tools: Gurobi, CVX, Python, MATLAB, R, C, Tableau
Languages: Korean (native), English (fluent)
Hobbies: Weightlifting, running